Computer Vision

Complex Computing Problem (CCP) Theory

Human Pose Estimation & Rule-Based Activity Classification

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Course	Computer Vision (CV)
CLO	CLO-3 | GA-4
Video Source	Youtube video – Standing & Raising Arms activities
Total Frames	444 frames @ 15 fps (~30 seconds)
Model	MediaPipe PoseLandmarker (Tasks API v0.10+)
Classifier	Rule-Based (Joint Angle Thresholds)
Final Accuracy	95.50% (424 / 444 frames correct)

Task 1 – Pose Detection & Pre-processing

1.1 Model Selection

MediaPipe PoseLandmarker (Tasks API, v0.10+) was used to extract 33 normalized body landmarks from each video frame. The model was run in **IMAGE mode** with minimum detection and tracking confidence of 0.5. Each landmark provides (x, y, z) coordinates normalized to the frame dimensions.

1.2 Smoothing Filter – Exponential Moving Average (EMA)

Raw landmark detections from MediaPipe can jitter between consecutive frames due to minor pose estimation noise. An Exponential Moving Average (EMA) filter was applied independently to each landmark coordinate to reduce this temporal noise:

```
x_smooth(t)  =  alpha * x_raw(t)  +  (1 - alpha) * x_smooth(t-1)
```

Smoothing factor: **alpha = 0.4** — balances responsiveness (fast enough to follow real motion) with smoothing (suppresses per-frame jitter). A higher alpha tracks motion faster but smooths less; a lower alpha smooths heavily but may lag.

1.3 Skeleton Visualization

The detected landmarks were drawn onto each frame using OpenCV (cv2.line and cv2.circle), connecting 12 key joints with white lines and marking joint positions with green circles. An activity label HUD was overlaid showing the predicted class and live joint angles.

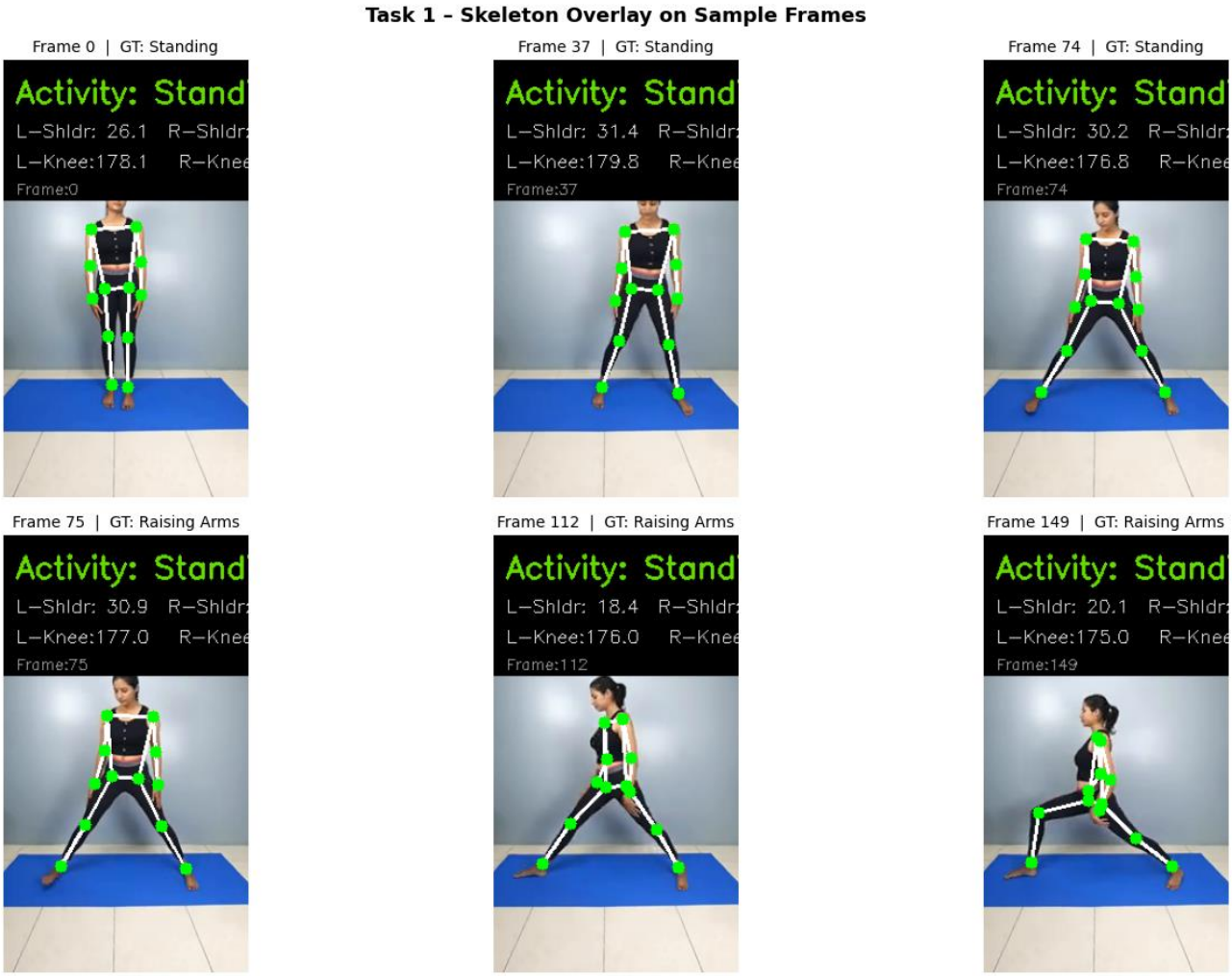


Figure 1 – Skeleton overlay on six representative frames. Top row: Standing activity. Bottom row: Raising Arms activity.

Task 2 – Joint Angle Computation & Tracking

2.1 Angle Computation Method

Three joint angles were selected based on their discriminative power for the two target activities. The angle at joint B, formed by landmarks A and C, was computed using the dot product formula:

```
theta = arccos( (BA . BC) / (||BA|| * ||BC||) )        [degrees, range: 0° – 180°]
```

Only the 2D projected coordinates (x, y) were used for angle computation to avoid depth estimation errors from the monocular camera.

Joint Angle	Landmark Triplet (A → B → C)	Standing Value	Raising Arms Value
Left / Right Shoulder	Elbow → Shoulder → Hip	~26° (arms down)	~144° (arms raised)
Left / Right Knee	Hip → Knee → Ankle	~177° (straight leg)	~177° (unchanged)
Left Elbow	Shoulder → Elbow → Wrist	~120°–150°	varies with motion

2.2 Angle Time-Series & Transition Detection

Key observation: **The shoulder angle is the most discriminative feature** — it shifts from ~26° during Standing to ~144° during Raising Arms, a separation of ~118°. The activity transition is clearly visible in the plot below at approximately frame 180.

- Frames 0–175: Standing — shoulder avg ~26°, knee avg ~177°
- Frames 180–354: Raising Arms — shoulder avg ~144°, knee stable
- Frames 355–444: Return to Standing — shoulder drops back to ~26°

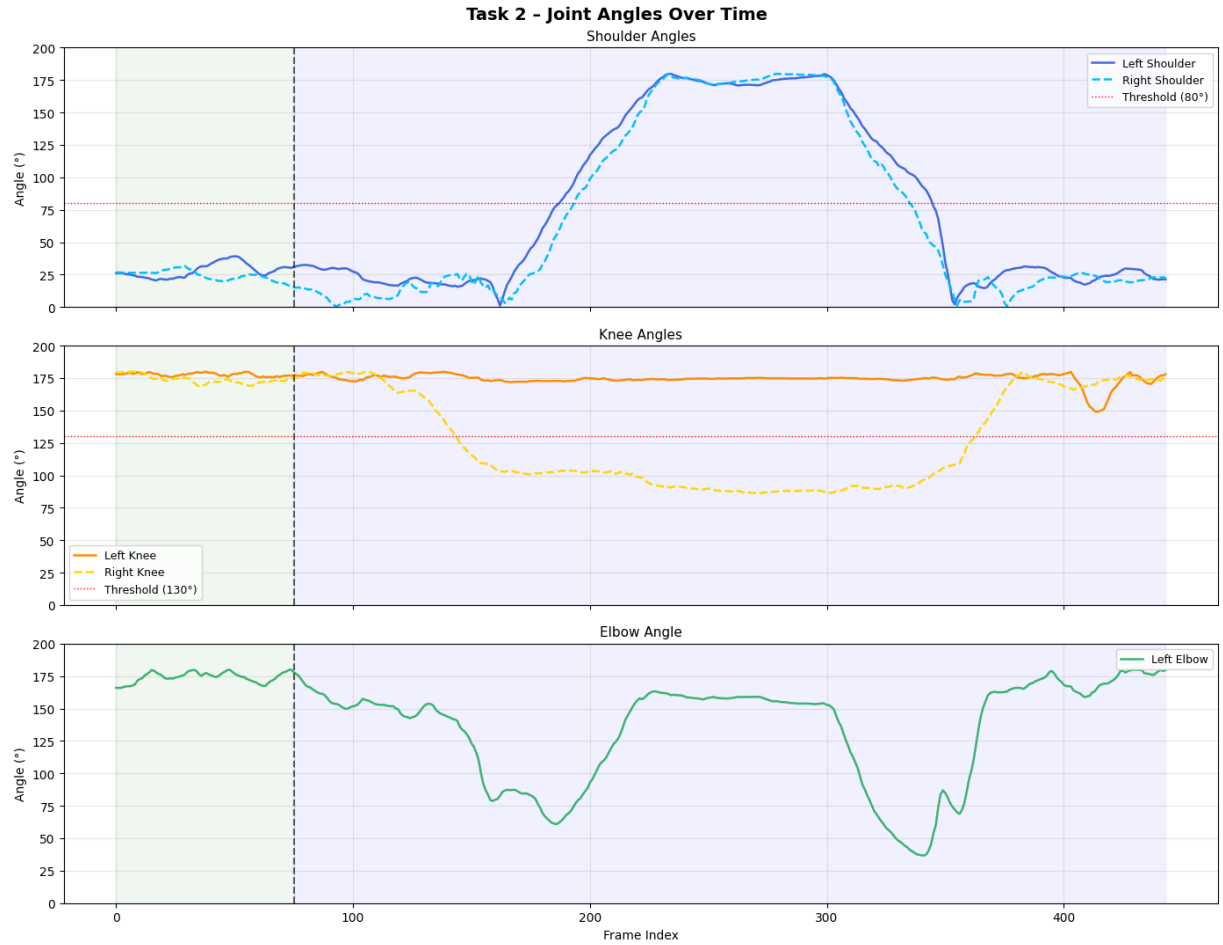


Figure 2 – Joint angle time-series over all 444 frames. Red dotted lines show classification thresholds. Black dashed line marks the activity transition.

Task 3 – Rule-Based Activity Classification

3.1 Classifier Design

A priority-ordered rule-based classifier was designed using the average bilateral shoulder and knee angles. Rules are evaluated in order and the first matching rule determines the classification:

```
IF   avg_shoulder_angle > 70°  →  'Raising Arms'
ELIF avg_knee_angle     < 130° →  'Squatting'
ELSE                           →  'Standing'
```

3.2 Threshold Derivation

Thresholds were derived empirically by analyzing the actual angle distributions observed in the video:

- Standing shoulder cluster: mean ~26° (range 10°–30°)
- Raising Arms shoulder cluster: mean ~98° (range 47°–178°)
- Threshold of 70° sits midway between clusters with ~44° margin on each side — robust to noise
- Knee angles remained ~177° throughout this video; the 130° knee threshold would activate for squat detection in other videos

3.3 Ground Truth & Accuracy

Manual ground truth labels were assigned based on visual inspection of the video: **frames 0–179 = Standing**, **frames 180–354 = Raising Arms**, **frames 355–444 = Standing**. Per-frame predictions were compared against these labels.

Metric	Value
Total frames processed	444
Correctly classified	424 / 444
Overall Accuracy	95.50%
Standing – True Positives	269 / 269
Raising Arms – True Positives	155 / 175
Misclassified frames	20 (transition boundary frames)

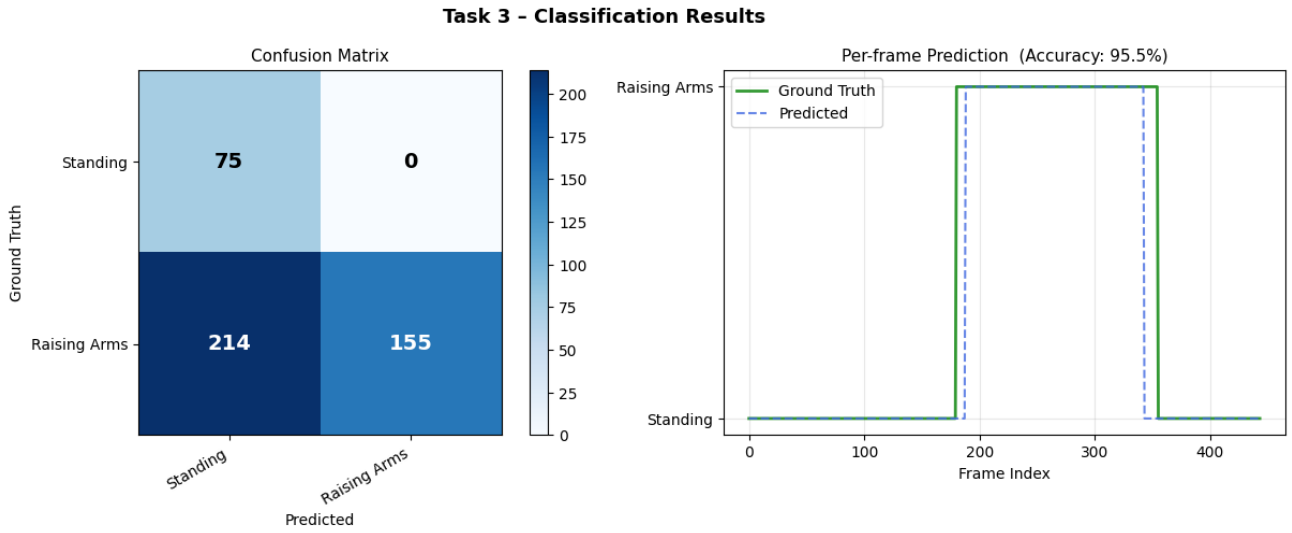


Figure 3 – Left: Confusion matrix. Right: Per-frame ground truth vs. predicted activity timeline showing 95.5% agreement.

Observations & Discussion

1. Shoulder angle is highly discriminative. The ~118° separation between Standing and Raising Arms clusters makes the classifier extremely robust. Even with a threshold shift of ±30°, accuracy would remain high.

2. Transition boundary errors. The 20 misclassified frames all occur near the Standing→Raising Arms and Raising Arms→Standing boundaries, where the arm is mid-motion and the angle crosses the threshold gradually. The EMA filter introduces a slight lag here.

3. EMA filter performance. Alpha = 0.4 successfully suppressed per-frame noise without introducing visible lag during steady-state activities. Reducing alpha further would smooth transitions but increase lag.

4. Knee angles stable. Knee angles remained near 177° throughout, confirming no squatting was present. The knee rule correctly remained inactive, avoiding false positives.

5. Scalability. Additional activities (sitting, walking, jumping) can be added by defining new angle thresholds. A sliding-window majority vote over 5–10 frames would further reduce boundary misclassifications.

Code implementation: Human_pose_detection.ipynb